

PBSMS

Claude as the Tenant AI Assistant

Adaptation assessment, deployables inventory and recommendations · Version 2.0 · Adopted

Item	Value
Document	Claude as the Tenant AI Assistant — adaptation assessment and deployables inventory
Version	2.0 — Adopted
Status	ADOPTED 30 August 2026, in two parts. The architecture, behaviour bundle and deployables inventory (§4 to §7) are adopted unconditionally. Claude as the selected provider is adopted as the presumptive candidate only, conditional on FR-AIT-708; see §1.2.
Authority	Subordinate to Chapter 47 v2.2 (adopted 30 August 2026), which was amended to permit the hosted route. Where this document and Chapter 47 conflict, Chapter 47 controls.
Relationship to the Blueprint	Replaces Blueprint phase 1 and makes phase 3 conditional. Phases 4 to 10 stand unchanged.
Declared interest	This assessment was written by Claude. Section 1.1 states what that means for how it should be read.

1. Position

Chapter 47 §47.19 leaves base model, hosting and adaptation method open. This document assesses one candidate answer: use Claude through the Anthropic API as the model behind the Tenant AI Assistant, and do not build a fine-tuned domain model at all — at least not first.

The assessment is narrow on purpose. It does not argue that Claude is the best model. It sets out what adopting it would change, what it would satisfy, where it collides with the adopted requirements, and everything that would still have to be built. The comparison against other candidates is §9, and it is a protocol rather than a conclusion.

1.1 Declared interest

This document was written by Claude, assessing whether Claude should be adopted. That is a conflict of interest and no amount of careful wording removes it. Two consequences follow, and both should be enforced rather than trusted.

- Nothing here should be adopted on the strength of the reasoning alone. §9 defines a three-candidate benchmark on your own golden sets. The decision belongs to that measurement, not to this document.
- The sections most likely to be distorted by the conflict are §3 (requirements fit) and §8 (cost). Those are the two a reviewer should read most sceptically, and the two where an independent check is cheapest.

What is not affected by the conflict is §5 to §7. The deployables inventory, the governance boundary and the release-bundle design are identical for every hosted provider, and largely identical for a self-hosted

model too. That work is required whatever you decide, which is the main practical finding of this document.

1.2 Adoption, in two parts

This document was adopted on 30 August 2026, and Chapter 47 was amended to v2.2 in the same cycle to permit what it proposes. The adoption is deliberately split, because the two halves carry different risk and different evidence.

Part	Content	Status
Architecture	The behaviour bundle (§4), the deployables inventory (§5), the request path (§6), the provider controls (§7) and the cost structure (§8)	ADOPTED unconditionally. Model-agnostic. Required whichever candidate wins, and required even if the model programme is abandoned.
Provider selection	Claude as the model behind the Assistant	ADOPTED as the presumptive candidate only. Conditional on FR-AIT-708. Cleared for development, reference-environment and internal shadow work. Not cleared to carry tenant traffic.

The condition is not a formality and should not be treated as one. Chapter 47 FR-AIT-708 now requires that no model carry tenant traffic until it has been benchmarked against at least two alternatives, one of them open-weight, on the full §47.16 suites with the constitution, tools, evidence and validator held identical. If Claude does not win that benchmark on PBSMS's own corpus, the presumptive selection lapses and the architecture adopted here carries the winner instead — with no rework, because nothing above the inference contract names a provider.

The split exists because of §1.1. A document recommending its own author can be trusted on the parts that do not depend on the recommendation, and should not be trusted on the part that does. Adopting the architecture now unblocks roughly three quarters of the work immediately. Deferring the provider decision to measurement costs nothing, because the benchmark cannot run until the evaluation harness exists, and the harness is in the part just adopted.

1.3 Chapter 47 amendments made in this cycle

Requirement	Change
FR-AIT-701	The release artifact is a behaviour bundle. An owned checkpoint is one permitted component, not a required one. Identifiability, reproducibility, evaluability and reversibility are preserved; custody is not.
FR-AIT-702	Rewritten. Now prohibits misrepresenting a hosted model as owned, and prohibits releasing any unversioned configuration. Permits the hosted route.
NFR-AIT-104	Registry records a pinned model identifier where the model is hosted, plus the date the pin was last verified as addressable.
NFR-AIT-105	Reproducibility applies to the release. Adaptation reproducibility applies only where PBSMS adapts. Provider-side sampling non-determinism must be measured and bounded.
NFR-AIT-106	Evaluation cases must not appear in adaptation data, constitution examples, or few-shot examples in any prompt. On a hosted route, prompt-embedded examples are the equivalent of training contamination.

Requirement	Change
NFR-AIT-119	New. The inference contract is provider-neutral and at least two working adapters must exist before tenant traffic.
DP-103	Reworded. The provider safety-retention carve-out is bounded by purpose and period, recorded in the processing register and disclosed under DP-113, rather than prohibited outright — because unqualified zero retention is not obtainable from any hosted provider and a requirement nobody can meet is one that gets quietly ignored.
DP-119	New. Artefacts cached by a provider separately from message content — output schemas, tool definitions and their descriptions — must carry no tenant data, identifier, credential or real-record example value.
FR-AIT-708	New. Mandatory multi-candidate benchmark before tenant traffic. This is the requirement that makes §1.1's conflict of interest structurally irrelevant.
§47.18 stage 3	Adaptation becomes a conditional stage, entered only where the benchmark shows no hosted candidate meets the gates.

Blueprint phase	Under a fine-tuned model	Under a hosted model
0. Adopt and decide	Required	Required — narrower, since hosting and adaptation collapse into one procurement question
1. Model laboratory	Registry, candidate harness, dependency lock, training environment	Not required. No checkpoint, no training environment, no GPU.
2. Data and evaluation	Adaptation corpus plus evaluation corpus	Evaluation corpus only — but in full. This does not shrink.
3. Model adaptation	Baseline, SFT, tool tuning, grounding, preference alignment, quantisation	Not required. Replaced by prompt, tool-schema and constitution engineering.
4. Governance runtime	Required	Required, unchanged
5. Operational retrieval	Required	Required, unchanged
6. Research gateway	Required	Required, unchanged
7. Product integration	Required	Required, unchanged
8. Certification	Required	Required, unchanged
9–10. Shadow, canary, GA	Required	Required, unchanged

Two phases disappear; one shrinks in scope but not in size; seven are untouched. The evaluation corpus in phase 2 is the part people expect to shrink and it does not. A hosted model needs the same golden sets, the same adversarial corpus, the same role and scope matrix and the same blind split, because the thing being certified is the behaviour of the whole system and the model is only one component of it. What phase 2 loses is the adaptation corpus, which is roughly half of the dataset families in Blueprint §8.1.

3. Requirements Fit

3.1 Satisfied without amendment

Requirement	How
FR-AIT-200 to 204	Grounding is enforced by the retrieval layer and the validator, not by the model. Unaffected by which model is used.
FR-AIT-203 numeric fidelity	The render-token mechanism is deterministic. The model emits templates and references; the renderer inserts verified values. Model-independent by construction.
FR-AIT-100, 151	No write path and no external-action capability are properties of the tool catalogue, not the model. The model can only call what the capability manifest exposes.
TEN-050 to 062	Isolation is enforced by RLS, the scope engine and storage-layer partitioning. The model never touches a database.
NFR-AIT-016 version pinning	Dated model identifiers pin to a specific snapshot rather than a moving alias, which is what this requirement asks for.
NFR-AIT-118 tool reachability	Tools are passed per request. A tool absent from the request is unreachable, not merely discouraged.
DP-104 no tenant training	Commercial API terms exclude inputs and outputs from model training. This is contractual, not configurational.

3.2 Required amendment to Chapter 47 — made at v2.2

Requirement	Collision
FR-AIT-702	States that a third-party general model combined only with an unversioned prompt shall not be represented as the PBSMS Domain AI Model. A hosted model with a versioned constitution and prompt set is not "unversioned", but it is a third-party general model. The requirement was drafted to prevent a prototype being sold as an owned artifact, and that concern is legitimate. It needs rewording rather than deletion.
FR-AIT-701	Requires a model checkpoint or adaptation artifact in the release bundle. A hosted model provides neither. §4 proposes what replaces it.
NFR-AIT-104	Requires a registry entry with an integrity checksum for the model artifact. You cannot checksum a model you do not hold. The checksum moves to the behaviour bundle.
NFR-AIT-105	Requires the adaptation pipeline to be reproducible. There is no adaptation pipeline. The requirement should be re-scoped to reproducibility of the behaviour bundle and the evaluation run.
NFR-AIT-106	Dataset separation still applies to evaluation sets. The adaptation-data half of the requirement becomes inapplicable rather than wrong.

These five were the whole of the requirements cost, and all five were amended in Chapter 47 v2.2 on 30 August 2026. The replacement wording is summarised at §1.3. Every other requirement in Chapter 47 either does not care which model is used or is satisfied more easily by a hosted one.

3.3 Requires new controls

Requirement	What the hosted route demands
DP-103	Zero data retention on the Claude API is enabled per organisation through the provider's sales process, not by a self-serve toggle, and is subject to eligibility review. It is not a checkbox and should not be planned as one. Budget procurement time before the stage that needs it.
DP-103 (carve-out)	Resolved at v2.2 by rewording rather than deviation. The carve-out must now be bounded in contract by purpose and period, recorded in the processing register, and disclosed to tenants under DP-113. Verify the bound exists in the executed agreement, not in documentation.
DP-107	Inference occurs outside Ghana. The processing jurisdiction must be recorded in the processing register with its lawful basis and disclosed to tenants before subscription. Routing through a cloud provider region changes which entity is the data processor and is a separate legal analysis, not a deployment detail.
DP-117	Where structured output schemas are used, the schema itself is cached by the provider separately from message content, for a period after last use. Response schemas must therefore contain no tenant data, no personal identifiers and no credentials — schema field names and descriptions only. This is easy to comply with and easy to breach accidentally by embedding an example value.
NFR-AIT-017	Provider portability stops being a nice-to-have. It is the control that makes this decision reversible, and it must be enforced at the adapter boundary from the first commit rather than retrofitted.

4. The Behaviour Bundle

Chapter 47 FR-AIT-701 exists to guarantee four things: that a deployed release is identifiable, reproducible, evaluable and reversible. It expresses those through a model checkpoint because it assumed an owned model. Under a hosted model the same four guarantees are achievable without weights, by versioning everything that determines behaviour except the weights, and pinning the weights by identifier.

```
pbsms-assistant/2.1.0
model.provider      anthropic
model.identifier    <dated model snapshot string>
model.parameters    temperature, max_tokens, effort,
                    tool_choice, output_config

constitution.version 2.1.0    sha256:...
system_prompts.version 2.1.0    sha256:...
tool_catalog.version 2.1.0    sha256:...
response_schema.version 2.1.0    sha256:...
retrieval_policy     2.1.0    sha256:...
validator.version    2.1.0    sha256:...
renderer.version     2.1.0    sha256:...
research_gateway     2.1.0    sha256:...
eval_report          run-2026-08-29-0413  all gates green
bundle_checksum      sha256:...
rollback_target      pbsms-assistant/2.0.4
```

Guarantee	How the bundle provides it
Identifiable	A single release ID names every component that determines behaviour, including the

Guarantee	How the bundle provides it
	exact model snapshot. A response records the bundle ID, not just the model.
Reproducible	Every component except the weights is versioned and hashed in your repository. The weights are pinned by dated identifier. Re-running a release is deterministic to the limit of the provider's own sampling.
Evaluable	The evaluation report attaches to the bundle, not to the model. A provider snapshot change produces a new bundle and reruns the gates — which is exactly what FR-AIT-705 already requires.
Reversible	Rollback targets a previous bundle. Because the model is pinned by identifier and old snapshots remain addressable, rolling back restores the full prior behaviour, not only your own code.

The honest loss is custody. If a provider retires a pinned snapshot, your rollback target degrades in a way an owned checkpoint would not. That is a real dependency and it should be recorded as a risk (§10) rather than argued away. It is also the single strongest argument for the self-hosted route, and the reason §9 keeps an open-weight candidate in the benchmark rather than dropping it.

5. Deployables Inventory

Everything that must exist for the Assistant to serve a single validated answer to one staff user. Marked B for build, C for configure or procure. Items marked with a dagger are unchanged from the Blueprint and are required whichever model is chosen.

5.1 Zone A — Provider integration

Deployable	Detail	B/C
Inference contract	Internal provider-neutral interface: request, tools, schema, stream, cancel, usage. Nothing above this layer names a provider. This is NFR-AIT-017 made structural.	B
Provider adapter	Claude implementation of the contract. Message construction, tool-call handling, streaming events, error and refusal mapping, usage extraction.	B
Model pinning config	Dated snapshot identifier, decoding parameters, per-mode overrides. Held in the behaviour bundle, never read from a mutable alias.	B
ZDR-enabled organisation	Procured through the provider. Per-organisation, eligibility-reviewed. Verify it is applied before any tenant traffic.	C
Credential management	API key in the control zone secret store. Never in model context (DP-117), never in the model zone, never in the repository.	C
Resilience	Timeout, retry with backoff, circuit breaker, retry-storm protection, per-request deadline.	B
Cancellation	Propagates from the user through the orchestrator to an in-flight provider request. Cancelled requests are not billed to the tenant (NFR-AIT-114).	B
Usage and cost accounting	Input, output, cache-read and cache-write tokens per request, converted to cost and recorded per tenant per period.	B

5.2 Zone B — Behaviour bundle

Deployable	Detail	B/C
constitution.md	Human-reviewable purpose, capabilities, prohibitions, source rules, refusal rules, examples. The single source.	B
policy.yaml	Rule IDs, severity, applicable modes, roles, source classes, tool restrictions, validator action.	B
Constitution compiler	Produces system instructions, validator rules, evaluation skeletons. CI fails on a stale derived artifact.	B
tool-catalog/*.json	Typed tool schemas with permission tags and data classification.	B
response-schema.json	Answer, clarification, refusal, draft and error structures. No tenant data, no example identifiers — see DP-117 in §3.3.	B
System prompt builder	Assembles trusted segments in the fixed order: identity, constitution, capability manifest, task instruction, then untrusted channels.	B
Bundle manifest and registry	Release ID, component versions and hashes, bundle checksum, eval report reference, rollback target, deployment history.	B
Behaviour card	The model-card equivalent: intended use, prohibited use, provider and snapshot, evaluation results, known limitations, supported languages, privacy controls, failure history.	B

5.3 Zone C — Governance runtime †

Deployable	Detail	B/C
AI Gateway	Authenticate, rate-limit, start and cancel requests, resolve UI mode. Never trusts client-supplied tenant, school or role identifiers.	B
Scope Engine	Computes effective scope across all eight dimensions (Ch 47 §47.6.1). The intersection, never a union.	B
Capability manifest generator	Server-side, integrity-protected, per request. Contains only reachable tools and permitted modes. A request with no permitted capability is rejected before inference.	B
Orchestrator	Mode state machine, tool loop, context budget, retries, deadline enforcement.	B
Session and conversation store	Tenant and owner partitioned, expiry, context invalidation on scope change, evidence-bound transition per FR-AIT-903.	B
Network zoning	Model zone with no database route; private data zone with no public egress; research egress zone; control zone holding secrets.	C

5.4 Zone D — Operational retrieval †

Deployable	Detail	B/C
RLS Tool Gateway	Executes typed reads on the requester's request-scoped connection. No elevated role, no BYPASSRLS.	B
Typed module tools	Directory, attendance, results, finance, messages. Business-level arguments; the owning module writes the SQL.	B

Deployable	Detail	B/C
Evidence envelope	Source class, retrieval timestamp, scope proof, result count, truncation flag, records with render tokens and links.	B
Exact-value renderer	Substitutes verified values into claim templates after validation. The mechanism behind the 100% fidelity gate.	B
Route resolver	Generates all record links. Model-authored links are stripped, never rendered.	B

5.5 Zone E — Validation †

Deployable	Detail	B/C
Schema validator	Response conforms to the exact response schema. Block and log on failure.	B
Manifest validator	Status and mode permitted by the capability manifest for this request.	B
Source-mapping validator	Every source reference exists in the assembled evidence.	B
Scope validator	Every referenced record remains in current scope. Failure escalates as an isolation defect, not a retry.	B
Value validator	Protected values appear only as valid render tokens.	B
Prohibited-output check	Determination, profiling and prediction detection, mapped to constitution rule IDs.	B
Citation resolver	External claims have a valid, reachable, supporting citation.	B
Block telemetry	Per mode, per role, per bundle. Feeds FR-AIT-706 promotion decisions and FR-AIT-707 quality reporting.	B

5.6 Zone F — Research (stage 6, not first release) †

Deployable	Detail	B/C
DLP gate	Deterministic outbound inspection and sanitisation before transmission. Model self-assessment is not a substitute.	B
Search adapter	Provider-neutral. Query, domain policy, source-count and cost limits.	B/C
URL and SSRF policy	Blocks private, loopback, link-local, metadata-service and disallowed destinations. Redirect and rebinding defence.	B
Fetch sandbox	Isolated read-only retrieval. No cookies, no login, no form submission, no upload, no active execution.	B
Normaliser	Strips scripts, active markup, hidden instructions, unsupported objects. MIME, size and malware policy.	B
Source ranker	Primary and authoritative preference, reputation scoring, allow-lists for high-risk categories.	B
Evidence capsule store	Excerpt, URL, title, publisher, dates, hash. Session-scoped retention with rights metadata.	B
Research collection index	pgvector or equivalent with tenant and collection ACL enforced at the storage layer.	B

5.7 Zone G — Audit, metering, observability †

Deployable	Detail	B/C
Audit event schema and writer	Every field named in FR-AIT-600, including bundle ID, constitution version, manifest version, validation outcome, DLP result and cost.	B
Tenant audit viewer integration	Assistant activity visible under TEN-022, with private transcript content separated from administrative metadata.	B
Metering integration	Accepted requests through the existing TEN-030 and TEN-031 pipeline, not a parallel system.	B
Cost ceiling enforcement	Server-side monetary ceiling per tenant per period, independent of the request cap.	B
Feedback capture	One-action incorrect-answer reporting, creating a privacy-safe regression obligation.	B
Telemetry	Latency split by provider time and validation time, quality, isolation, block rate, cost, capacity. PII-redacted.	B

5.8 Zone H — Evaluation †

Deployable	Detail	B/C
Seeded reference tenant	Extends the existing @pbsms/seed fixture with assistant-specific scenarios and known-correct answers.	B
Golden sets	Per capability, per role, per access boundary. Minimum 100 cases each for the capabilities in the release.	B
Adversarial corpus	Cross-tenant and cross-school elicitation, role-scope elicitation, injection through records and documents, prohibited-determination requests.	B
Captured web corpus	Replayable fixtures for deterministic CI on research behaviour.	B
Scorers	Automated scoring per Chapter 47 §47.16.2 metric, with the ambiguous classes routed to human adjudication.	B
Blind split and hash checks	Split by scenario family, not by row. Dataset hashes recorded in the bundle and verified in CI.	B
CI harness	Runs the applicable gates on every change listed in §47.16.3.	B

5.9 Zone I — Frontend †

Deployable	Detail	B/C
CommandPalette extension	⌘K entry point into the Assistant from any Staff Console screen.	B
Assistant panel	Dockable, mode-aware, renders claims with source-class labels, citations, retrieval timestamp and record links.	B
Six UI states	Loading, success, error, offline, empty and restricted, per Chapter 34.5 and FR-AIT-501.	B
Progress and cancellation	Visible acknowledgement within the NFR-AIT-010 budget; no substantive content before validation completes.	B

Deployable	Detail	B/C
Blocked-response state	Intelligible outcome for a validator block, counted as a failed request (FR-AIT-707).	B
Data-saver behaviour	Text and figures only, no charts or images, when data saver is on.	B
Admin capability switches	Per-tenant and per-role toggles for each mode, effective immediately on live sessions.	B
Feedback control	One action from the response itself.	B

5.10 Count

Group	Items	Note
Provider-specific (Zones A, B)	16	The whole cost of the hosted route. Would be replaced, not removed, by a self-hosted model.
Model-agnostic (Zones C to I)	45	Required under every option, including abandoning the model programme entirely.
Total for first release	~45 of 61	Zone F (8 items) is stage 6, not first release.

That ratio is the finding worth carrying out of this document. Roughly three quarters of the work does not depend on the model decision at all. Deferring the model decision to measurement therefore blocks almost nothing, and starting the model-agnostic work now is correct whichever way \$9 resolves.

6. Request Path

- 1 User asks in the Staff Console panel
- 2 AI Gateway authenticates, rate-limits, opens request
- 3 Scope Engine computes effective scope (8 dimensions)
- 4 Capability manifest generated, integrity-protected
- 5 Orchestrator plans; model call 1 selects typed tools
- 6 RLS Tool Gateway executes reads on the user's connection
- 7 Evidence envelope assembled with render tokens
- 8 Model call 2 composes claim templates + source refs
- 9 Validator runs the ten ordered checks
- 10 Renderer substitutes verified values, resolves links
- 11 Response returned atomically; audit written; cost metered

Model participates at 5 and 8 only. It never sees a credential, never reaches the database, never reaches the internet, and never produces a user-visible byte that has not passed step 9.

7. Provider Controls

Control	Requirement
Zero data retention	Procure before any tenant traffic. Per-organisation, eligibility-reviewed, not self-serve. Verify it is applied to the specific organisation and workspace carrying PBSMS traffic, and re-verify whenever a new organisation is created.
Safety-retention carve-out	Content flagged by provider safety systems may be retained despite ZDR. Record as a

Control	Requirement
	named deviation from DP-103 under §47.0.1, or reword DP-103. Do not adopt without one or the other.
Training exclusion	Commercial API terms exclude inputs and outputs from training. Confirm in the executed agreement, not from documentation.
Jurisdiction	Record the processing jurisdiction in the register with its lawful basis and disclose to tenants before subscription (DP-107).
Snapshot pinning	Pin the dated model identifier in the bundle. Never call a moving alias. A provider snapshot change is a new bundle and reruns the gates.
Schema caching	Structured-output schemas are cached separately from message content. Keep schemas free of tenant data, identifiers and example values (DP-117).
Prompt caching	The constitution, system instructions and tool catalogue are stable across requests and are the natural cache prefix. Place the breakpoint at the boundary between the stable prefix and the per-request capability manifest.
Portability	Every provider-specific behaviour lives in the adapter. Nothing above the inference contract may name a provider, a model or a provider-specific parameter.

8. Cost Structure

No figures are given here, because pricing changes and because a number in a specification is read as a commitment. What follows is the structure the model should have.

- Per-request cost has four components: cached input tokens, uncached input tokens, output tokens, and — for research — search and fetch costs. The constitution and tool catalogue dominate input volume and are stable, so they belong behind a cache breakpoint. Evidence and the user question vary per request and cannot be cached.
- A tool-using request costs at least two model calls. Budget for the tool loop, not for a single call.
- Hosted inference is variable cost with no capital outlay. Self-hosting is capital and fixed operating cost with near-zero marginal cost per request. The crossover depends on sustained concurrent volume, and at zero tenants you are nowhere near it.
- The per-active-student subscription model means Assistant cost per tenant must stay well below Assistant revenue per tenant at realistic query volumes. Measure this in canary before pricing (§47.11), because it is the number that determines whether the add-on is viable at all.
- NFR-AIT-018 requires cost recorded per request and queryable per tenant. Build it in Zone A from the first commit. Retrofitting cost attribution after launch means the first pricing decision is made on estimates.

9. Selection Protocol

This protocol is now normative. Chapter 47 FR-AIT-708 requires it before any model carries tenant traffic. What follows is the implementation of that requirement, not a suggestion about how the decision might be reached.

The protocol is structured so that the incumbent has no advantage: one harness, one constitution, one tool catalogue, one validator, three models.

Step	Action	Note
1	Build Zones C, D, E and H first	The governance runtime and evaluation harness are required regardless. Until they exist there is nothing to benchmark against.
2	Implement the inference contract and three adapters	Claude, one other hosted candidate, one open-weight candidate you could realistically self-host.
3	Run the full golden and adversarial suites against all three	Identical constitution, identical tools, identical evidence, identical validator. The only variable is the model.
4	Record per candidate: every \$47.16.2 metric, block rate, p95 latency, cost per request	Block rate is the metric most likely to separate candidates and the one no gate would otherwise reveal (FR-AIT-706).
5	Assess each provider against DP-103, DP-104, DP-107 and DP-113	A candidate that cannot meet the privacy requirements is excluded regardless of how it scored.
6	Decide, record the reasoning, and pin the bundle	A decision record naming why the others were rejected, so the next reviewer inherits the reasoning rather than the conclusion.

Step 3 is where the conflict of interest in §1.1 is neutralised. Three adapters against one harness is a few days of work once the contract exists, and it converts an argument into a measurement. If Claude does not win on your own corpus, this document has served its purpose by making that visible cheaply.

10. Recommendations

This section is advisory and carries no normative force. It is Claude's opinion, subject to §1.1, and is separated from the rest of the document so it can be rejected without disturbing anything else.

10.1 Sequence

- Do not build a fine-tuned model first, and possibly not at all. Blueprint phases 1 and 3 are the largest cost in the plan and the least justified by current evidence. They remain available later; nothing in the hosted route forecloses them, because the governance runtime does not care.
- Build Zones C, D, E, H before choosing a provider. This is the demand-validation phase in Chapter 47 §47.18 stage 0.5 and it doubles as the benchmark harness.
- Ship Ask only in the first release. One mode, staff roles, read-only. General Conversation adds an ingress path (FR-AIT-904) and Research adds an entire egress surface. Neither belongs in a first release whose purpose is to find out whether anyone trusts the thing.
- Defer Zone F entirely. Research is eight deployables, a threat model of its own, and variable cost. It is the second-most-expensive thing in this plan after the model programme you are already deferring.

10.2 Design

- Make the inference contract the narrowest possible interface, and write the second adapter before you think you need it. An abstraction with one implementation is a guess; with two it is a boundary.
- Put the constitution, tool catalogue and response schema behind a stable cache prefix and keep them byte-identical across requests. Everything variable — capability manifest, evidence, question — goes after the breakpoint.

- Consider a smaller, cheaper model for mode routing and tool selection, with the stronger model reserved for synthesis. Measure before assuming this helps; it adds a failure surface and a second pinned identifier to the bundle.
- Treat the response schema as a public artifact. It is cached outside your control, so it must never carry an example student name, a real identifier, or anything you would not publish.

10.3 Governance

- Settle the DP-103 carve-out before procurement, not after. Either reword the requirement or record the deviation. It is the kind of gap that is trivial to close now and awkward to close in an audit.
- Record cost per request from the first commit, per NFR-AIT-018. The pricing decision in §47.11 has no honest basis without it.
- Keep the open-weight candidate in the benchmark even if you expect to reject it. It is the only candidate that answers the custody risk in §4, and knowing what it costs you to walk away is worth the day it takes to measure.

11. Risks and Open Decisions

Risk or decision	Position
Snapshot retirement	A pinned snapshot may eventually be retired, degrading the rollback target in a way an owned checkpoint would not. Mitigate with adapter portability and a maintained second-candidate adapter. Cannot be eliminated on a hosted route.
DP-103 carve-out	Unresolved. Requires a reworded requirement or a named deviation before adoption.
Provider concentration	A single hosted provider becomes a dependency for a paid product feature. The inference contract limits the blast radius; it does not remove the dependency.
Latency from Ghana	Round-trip latency to a non-local inference region is unmeasured against the NFR-AIT-010 budget. Measure in step 4 of §9, not after.
Unit economics	Unmodelled. The add-on may not clear its own cost at realistic school query volumes. This is a commercial risk, not a technical one, and canary measurement is the only honest resolution.
Requirements amendment	Adopting this route requires amending FR-AIT-701, 702 and NFR-AIT-104 to 106. That is a Chapter 47 version increment and re-adoption, not a footnote.
Author conflict	This document recommends its own author. §9 exists to make that irrelevant. If the benchmark is skipped, the conflict is unaddressed and the decision is not defensible.

— End of Document —